

YUNCHENG, China

WMO: 539590

Lat: 35.1118N

Lon: 111.0670E

Elev: 1235

StdP: 14.05

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.4	(c) 19.8	(d) -8.7	(e) 3.6	(f) 30.0	(g) -3.5	(h) 4.8	(i) 30.9	(j) 18.7	(k) 36.0	(l) 16.4	(m) 36.7	(n) 3.0	(o) 20	(p) 0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.7	(c) 97.8	(d) 72.2	(e) 95.3	(f) 72.2	(g) 92.8	(h) 71.7	(i) 78.3	(j) 90.2	(k) 77.1	(l) 89.0	(m) 75.9	(n) 87.3	(o) 6.8	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	138.1	85.4	73.7	131.4	84.3	72.4	125.6	83.2	43.1	90.6	41.7	89.1	40.5	87.6	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.4	(b) 15.6	(c) 13.2	(e) 11.3	(f) 102.6	(g) 3.3	(h) 2.0	(i) 8.9	(j) 104.0	(k) 7.0	(l) 105.2	(m) 5.1	(n) 106.4	(o) 2.7	(p) 107.9		
(a) 18.4	(b) 15.6	(c) 13.2	(e) 9.1	(f) 79.8	(g) 3.3	(h) 1.6	(i) 6.8	(j) 81.0	(k) 4.9	(l) 81.9	(m) 3.0	(n) 82.8	(o) 0.6	(p) 84.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.0	31.7	39.2	50.7	62.1	71.2	79.3	82.5	79.9	70.7	59.4	45.6	34.4	(7)	(8)
	DBStd	18.63	5.20	6.77	7.76	7.42	6.13	5.71	5.30	5.85	6.47	6.33	6.60	5.12		
	HDD50	1617	568	305	87	6	0	0	0	0	0	5	161	485		
	HDD65	4101	1033	722	447	138	16	1	0	0	18	195	581	950		
	CDD50	4897	0	4	108	368	658	879	1007	926	621	295	31	0		
	CDD65	1905	0	0	3	51	209	430	542	461	189	20	0	0		
	CDH74	20654	0	0	50	554	2099	4928	6520	4897	1459	147	0	0		
	CDH80	9359	0	0	8	157	811	2417	3272	2194	477	23	0	0		
	WSAvg	5.3	4.5	5.5	6.1	6.1	6.0	5.7	5.9	5.7	5.1	4.5	4.3	4.4		
	PrecAvg	20.0	0.2	0.4	0.7	1.4	2.1	2.2	3.9	3.1	3.2	1.9	0.9	0.2		
(15) Precipitation	PrecMax	33.2	0.6	0.7	1.7	4.4	5.5	4.8	10.6	7.2	11.0	4.8	3.4	0.6	(16)	(17)
	PrecMin	12.0	0.0	0.0	0.0	0.1	0.4	0.7	1.1	0.3	0.6	0.2	0.0	0.0		
	PrecStd	4.4	0.2	0.2	0.5	0.9	1.3	1.1	2.0	1.7	2.3	1.2	0.9	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	63.9	79.5	90.2	95.8	100.4	101.5	98.9	94.7	83.0	68.5	53.1	(20)	(21)
		MCWB	38.1	46.7	56.5	63.4	64.4	69.8	74.6	74.9	71.8	61.3	52.2	40.1		
	2%	DB	47.8	58.8	73.7	84.3	91.2	97.2	97.8	95.7	89.0	78.0	64.2	49.6		
		MCWB	36.3	44.1	53.2	61.0	64.3	68.7	74.3	74.2	69.5	60.8	50.5	38.7		
	5%	DB	44.9	55.1	69.2	80.3	88.0	94.4	95.3	92.8	85.2	74.2	60.8	46.7		
		MCWB	34.7	42.0	51.1	58.9	63.2	68.7	74.7	73.9	67.9	59.3	49.5	36.9		
	10%	DB	42.0	51.5	64.8	76.3	84.7	91.4	92.6	89.9	81.9	70.7	57.3	44.1		
		MCWB	33.1	40.2	49.0	57.5	62.6	68.2	74.4	73.5	66.8	58.1	48.0	35.5		
	0.4%	WB	39.7	49.0	59.3	68.4	69.9	76.6	80.6	79.4	76.8	66.1	55.3	43.7	(27)	(28)
		MCDB	47.8	61.3	72.8	81.6	84.5	90.5	93.5	90.3	88.7	75.1	62.6	48.3		
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.6	45.5	55.1	64.1	68.0	74.3	79.0	77.9	73.1	63.8	53.0	40.1	(30)	(31)
		MCDB	45.5	55.5	70.5	77.8	82.4	88.0	91.1	89.3	84.2	72.0	60.5	46.7		
	5%	WB	35.6	43.6	52.5	61.5	66.7	72.5	77.7	76.7	70.8	62.0	51.1	38.3		
		MCDB	44.1	52.9	66.0	76.1	80.5	86.1	89.9	87.8	80.1	69.8	58.1	44.6		
	10%	WB	33.6	41.3	50.2	59.3	65.3	71.1	76.5	75.6	68.9	60.2	49.1	36.5		
		MCDB	41.3	50.3	62.8	72.4	78.9	84.4	88.5	86.4	77.9	67.7	55.9	42.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	18.0	19.4	20.4	20.6	19.1	15.7	15.0	15.6	16.6	17.0	16.5	(36)	(37)
		MCDBR	22.3	24.4	27.1	27.6	27.7	24.7	19.6	18.5	21.1	23.0	22.4	20.9		
		MCWBR	13.4	13.1	12.4	10.6	8.8	7.1	5.3	5.1	7.3	9.9	11.5	12.2		
	5% WB	MDBR	20.1	21.5	23.8	22.2	20.4	18.6	16.1	15.2	16.3	17.4	18.1	17.6		
		MCDBR	12.4	12.1	11.4	10.1	8.1	6.6	5.6	5.3	7.1	8.5	10.3	11.0		
		MCWBR														
(40) Clear-Sky Solar Irradiance	taub		0.518	0.606	0.616	0.623	0.623	0.696	0.710	0.735	0.666	0.647	0.564	0.493	(41)	(42)
	taud		1.739	1.601	1.613	1.636	1.689	1.592	1.615	1.550	1.643	1.617	1.708	1.806		
	Ebn at Noon		199	196	212	220	223	207	203	193	198	184	184	197		
	Edh at Noon		57	73	78	80	77	85	82	86	75	71	58	50		
(44) All-Sky Solar Radiation	RadAvg		810	972	1325	1591	1731	1729	1572	1417	1152	1007	832	759	(45)	(46)
	RadStd		89	165	137	136	134	156	172	173	118	148	122	95		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-1.49	-3.17	N/A	-1.39	-1.62	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page